EV TEAM PROCEDURES

Relevant to: EV Teams

NASA SUITS
2025-26 Challenge

NASA SUITS EVA Procedures

Last Updated: 3/2/2026

Updated By: Tech Team

This document **may** be updated ahead of test week in May 2026.

Notes:

This document will likely undergo several more revisions as we do dry runs in preparation for test week. NASA SUITS aims to have each test session run for 45 minutes. This allows time on each end of the window to set up and to transition from group to group. Therefore, the PR and Spacesuit portions will each take about 25 minutes with a few cutoffs. The reasoning for the cutoffs is to prevent teams from getting stuck and allowing other functionalities to be showcased, i.e. "Something isn't working, and we are behind, so in order to see your other functionalities, let's move on."

Change Log:

1. Modify PR Team timeline to be more real-world accurate. Add verbal aspects to checklists for design evaluators to confirm procedures, add cover page. -RE 2-17-26
2. Separate PR and EV team procedures into separate documents. -RE 2-18-26
3. Alter procedures to be more accurate to Test Week following dry run, change wording to be uniform throughout -GC, LC, RE 2-27-26
4. Add procedures for handling errors that may occur -LC, RE 3-2-26

General Test Timeline

Both the rover and EV teams will begin their procedures at the same time. The following procedures are done concurrently with the PR team and their procedures but are **NOT** dependent on one another. Cutoff times are fluid for each station, but 45 min is hard stop.

EV	Ideal Start Time	Time to complete
EVA Egress • Connect UIA to DCU • Start Depress • Prep O2 Tanks • End Depress, Check Switches and Disconnect	0 min	7 min
Navigate to LTV location	7 min	3 min
LTV Repair • Complete system scan and restart • Identify and repair issues with the LTV	10 min	10 min
Navigate to PR location	20 min	3 min
Ingress • Connect UIA to DCU • EMU config • Disconnect	23 min	2 min
EVA completion	25 min	N/A

Procedures

CAPCOM:

- *Reset all UIA switches to Down Position*
- *Reset all DCU switches to Back Position*
- *Reset LTV Task Board to pre-repair settings*
- *START EVA Telemetry (at EVA start time)*
- *Announce scenario start, give go to begin*

EVA

CAPCOM:

- *Monitor UIA Switches*

Connect UIA to DCU and start Depress (1 min)

1. UIA and DCU: EV1 verify umbilical connection from UIA to DCU
2. UIA: EV1 - EMU PWR – ON
3. DCU: BATT – UMB
4. UIA: DEPRESS PUMP – ON

Prep O2 Tanks (3 min)

1. UIA: OXYGEN O2 VENT – OPEN
2. HMD: Wait until both Primary and Secondary OXY tanks are < 10psi
3. UIA: OXYGEN O2 VENT – CLOSE
4. DCU: OXY – PRI
5. UIA: OXYGEN EMU-1 – OPEN
6. HMD: Wait until EV1 Primary O2 tank > 2950 psi
7. UIA: OXYGEN EMU-1 – CLOSE
8. DCU: OXY – SEC
9. UIA: OXYGEN EMU-1 – OPEN
10. HMD: Wait until EV1 Secondary O2 tank > 2950 psi
11. UIA: OXYGEN EMU-1 – CLOSE
12. DCU: OXY – PRI

Prep Coolant Tank (1 min)

1. DCU: PUMP – OPEN (Fill water tanks)
2. UIA: EV-1 SUPPLY WATER – OPEN
3. HMD: Wait until water EV1 Coolant Storage is > 95%
4. UIA: EV-1, SUPPLY WATER – CLOSE

END Depress, Check Switches and Disconnect (2 min)

1. HMD: Wait until SUIT Pressure and O2 Pressure = 4
2. UIA: DEPRESS PUMP PWR – OFF
3. DCU: BATT - PRI
4. DCU: BATT – LOCAL
5. UIA: EV-1 EMU PWR – OFF
6. DCU: FAN – PRI
7. DCU: PUMP – CLOSE
8. DCU: CO2 - PRI
9. DCU: Verify OXY – PRI
10. EV-1 disconnect UIA and DCU umbilical
11. Verbally announce completion of egress
12. Begin navigation procedure

Navigate to LTV Worksite (3 min)

1. Drop pin and determine best path to reach the LTV
2. Verbally confirm the path has been generated
3. Exit airlock and begin navigation to LTV worksite
    a. *Teams can create custom procedures to showcase additional navigation features and use of AI assistant*
4. Upon arrival at worksite, verbally confirm safe access to LTV

LTV Repair (10 min)

NOTE: ltv-repair-procedures.pdf contains example procedures teams may expect to see. Actual procedures will be sourced from the TSS during Test Week. All instructions and relevant information will be included in ltv-repair-procedures.pdf

CAPCOM:

- *Monitor LTV errors section in CAPCOM*

1. Verbally announce the beginning LTV diagnosis
2. Utilize AI assistant to retrieve the Exit Recovery Mode (ERM) procedure
3. Perform the ERM procedure and announce completion
4. Utilize AI assistant to retrieve the NAV Restart procedure
5. Perform the NAV Restart procedure and announce completion

6. Utilize AI assistant to perform LTV diagnosis
    a. *Upon analysis, the AI assistant should retrieve the relevant repair procedures from the TSS*
7. Based on procedures provided by AI assistant, perform the necessary operations on the LTV until all errors are resolved
    a. *Procedures should be relayed either verbally or visually through the HMD by the AI assistant*
8. Verbally announce successful repair of LTV
    a. “LTV repairs successful, returning to PR”

Navigate to PR Location (3 min)

1. Drop pin and determine optimal path to reach the PR
2. Verbally confirm path has been generated
    a. “Path generated, beginning navigation to PR”
3. Begin navigation to the PR
4. Upon arrival at the PR, announce arrival and begin ingress

EVA Ingress (2 min)

5. UIA and DCU: EV1 connect UIA and DCU umbilical
6. UIA: EV-1 EMU PWR – ON
7. DCU: BATT – UMB
8. UIA: OXYGEN O2 VENT – OPEN (Vent O2 tanks)
9. HMD: Wait until both Primary and Secondary OXY tanks are < 10psi
10. UIA: OXYGEN O2 VENT – CLOSE
11. DCU: PUMP – OPEN (Empty water tanks)
12. UIA: EV-1 WASTE WATER – OPEN
13. HMD: Wait until water EV1 Coolant tank is < 5%
14. UIA: EV-1, WASTE WATER – CLOSE
15. UIA: EV-1 EMU PWR – OFF
16. DCU: EV-1 disconnect umbilical

CAPCOM

- *Stop EV Telemetry*
- *Prep for next team*
    - *Reset PR to home base*
    - *Reset LTV errors, and physical task board*

- *Verify DCU and UIA are in default state*

Handling Error Scenarios

If an error occurs, the relevant procedures may be used to mitigate risk or abort the EVA. Please note that the occurrence of one or more of the following errors should only be used to test your design's response and will NOT prematurely end your test.

Off Nominal Heart Rate: Exceeding Nominal Range

1. Announce heart rate exceeds nominal range, reduce physical activity
   a. "Heart rate too high, slowing down temporarily"
2. Control breathing and monitor heart rate until nominal
3. Announce when heart rate is nominal, resume activity
   a. "Heart rate back in range"

Off Nominal Suit Oxygen Pressure

1. DCU: OXY – SEC
   a. Alert the astronaut to return to PR as soon as possible

Off Nominal Suit CO2 Pressure: Exceeding Nominal Range

1. If DCU: CO2 – PRI, DCU: CO2 – SEC; If DCU: CO2 – SEC, DCU: CO2 – PRI

Off Nominal Suit Pressure Other: Exceeding Nominal Range

a. If after decompress sequence, alert the user to return to PR as soon as possible

Off Nominal Total Suit Pressure

1. Check oxygen tank pressure
2. If oxygen tank value deviates from the nominal range, perform Off Nominal Suit Oxygen Pressure procedure
3. Check CO2 scrubber value
4. If CO2 scrubber value deviates from the nominal range, perform Off Nominal Suit CO2 Pressure procedure

Off Nominal Helmet CO2 Pressure: Exceeding Nominal Range

1. DCU: CO2 – SEC

a. *Alert the astronaut to return to PR as soon as possible*

Off Nominal Primary Fan: RPM Above or Below 30,000

1. DCU: FAN – SEC
    b. *Alert the astronaut to return to PR as soon as possible*

Off Nominal CO2 Scrubber: Filled Beyond 60% Capacity

1. If DCU: CO2 – PRI, DCU: CO2 – SEC; If DCU: CO2 – SEC, DCU: CO2 – PRI

Off Nominal Temperature: Exceeding Nominal Range

1. Announce temperature exceeds nominal range, reduce physical activity
    b. "Temperature too high, slowing down temporarily"
4. Control breathing and monitor temperature values until nominal
5. Announce when temperature is nominal, resume activity
    a. "Heart rate back in range"

Off Nominal Battery: Below Nominal Range: Less than 20% Capacity

1. If DCU: BATT – PRI, DCU: BATT – SEC